

APPENDIX 3.1
(referred to in paragraphs 3.18 and 3.80)

Insulation measures

There are three main measures of how well a given leaf material will perform as an insulator: thermal conductivity, thermal resistance, and thermal transmittance. These are defined below.

Thermal conductivity (K-value)

The thermal conductivity of a material is a measure of the rate at which the given material will pass heat.

It is expressed in units of watts per metre per degree of temperature (W/m K).

Thermal resistance (R)

The thermal resistance of a material may be calculated by dividing the thickness of the material (in metres) by the thermal conductivity of the material.

Thermal resistance is expressed in units of $\text{m}^2 \text{ K/W}$

Note:

Accepted resistance of 50mm cavity = $0.18 \text{ m}^2 \text{ K/W}$

Accepted resistance of external air surface = $0.06 \text{ m}^2 \text{ K/W}$

Accepted resistance of internal air surface = $0.12 \text{ m}^2 \text{ K/W}$

Thermal transmittance (U-Value)

Thermal transmittance is a measure of how much heat will pass through 1m^2 of a structure when the air temperatures on either side differ by one degree Kelvin. It is a measure of the thermal efficiency of a building element (ie wall, floor, roof, whole building etc). It is the reciprocal of Thermal Resistance (R).

U-values are expressed in units of watts per m^2 per degree of temperature difference ($\text{W/m}^2\text{K}$).